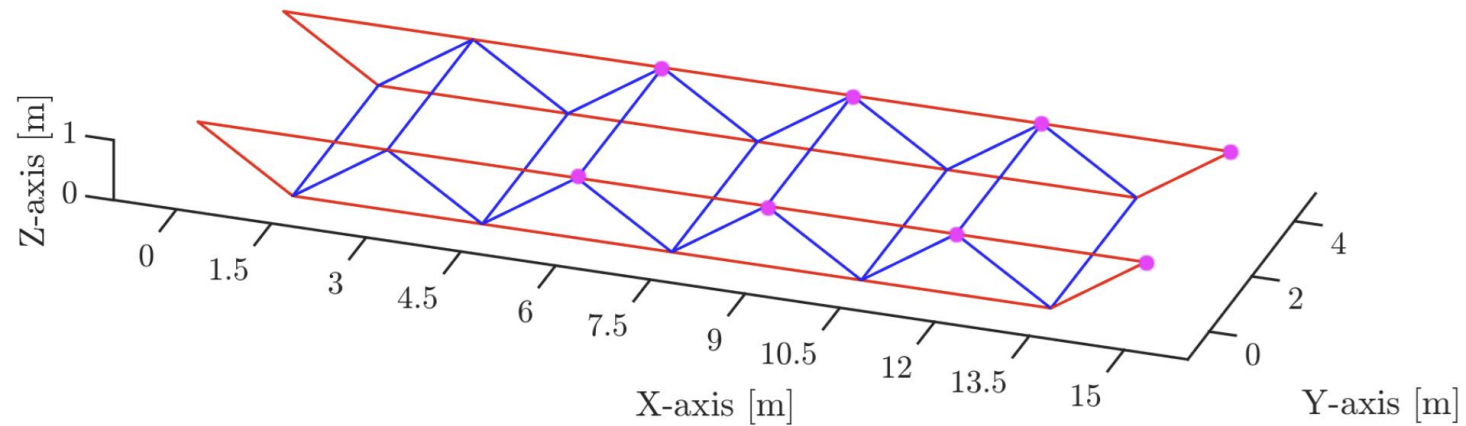


Analysis of the dynamic behavior of a steel canopy

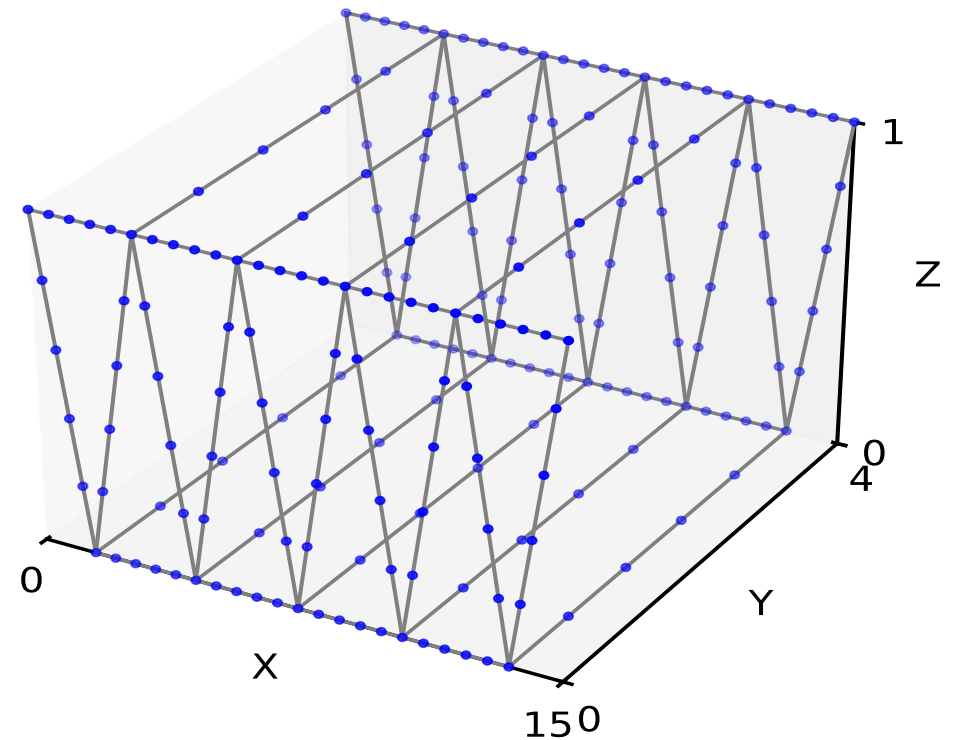
Group 16

Tom Beauve, Lilian Steimetz



Modeling strategy

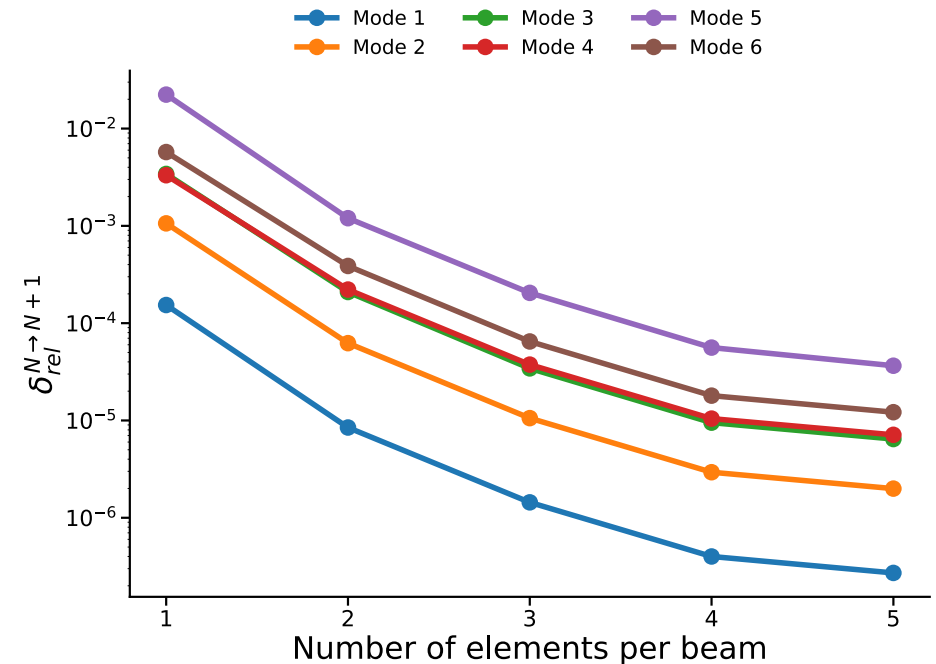
- Equal (variable) number of elements per beam
- NX model : Timochenko beam elements
- Python model : Euler-Bernoulli beam elements
(shear neglected)
- Total mass (using RBM in translation) : 1681 kg
- Equal mass for both models



Python model with 5 elements per beam

Modal analysis : convergence & comparison of the 2 models

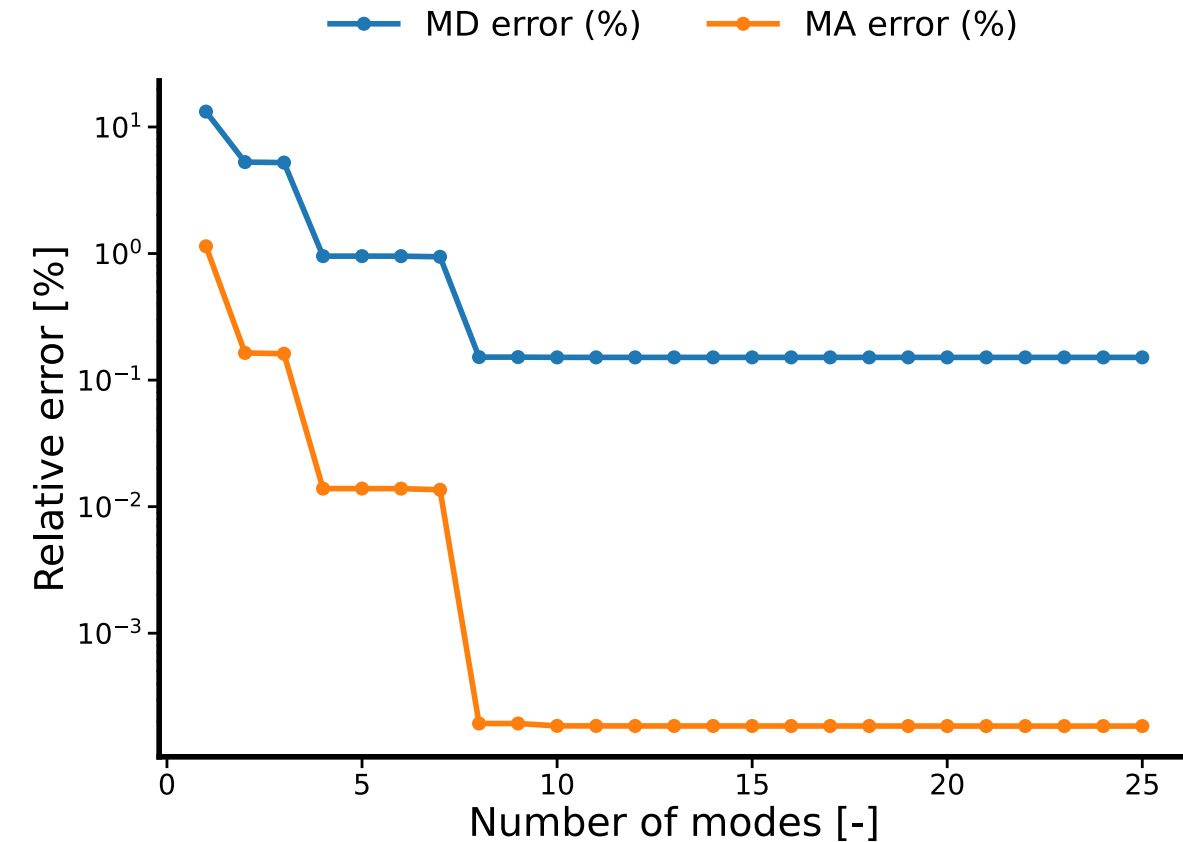
- Convergence after 5 elements per beam
- Python model consistently stiffer than NX (Euler-Bernoulli vs Timochenko)



Natural frequencies for 10 elements per beam :

Model	Mode 1	Mode 2	Mode 3	Mode 4	Mode 5	Mode 6
Python (Hz)	2.573	6.844	10.856	12.916	13.866	14.877
NX (Hz)	2.569	6.827	10.816	12.864	13.856	14.816
Rel. Error (%)	0.14	0.25	0.36	0.40	0.07	0.41

Stationary response : MD and MA comparison



Spatial factor :

$$X_s = \frac{\mathbf{x}_s^T \mathbf{g}}{\mu_s}$$

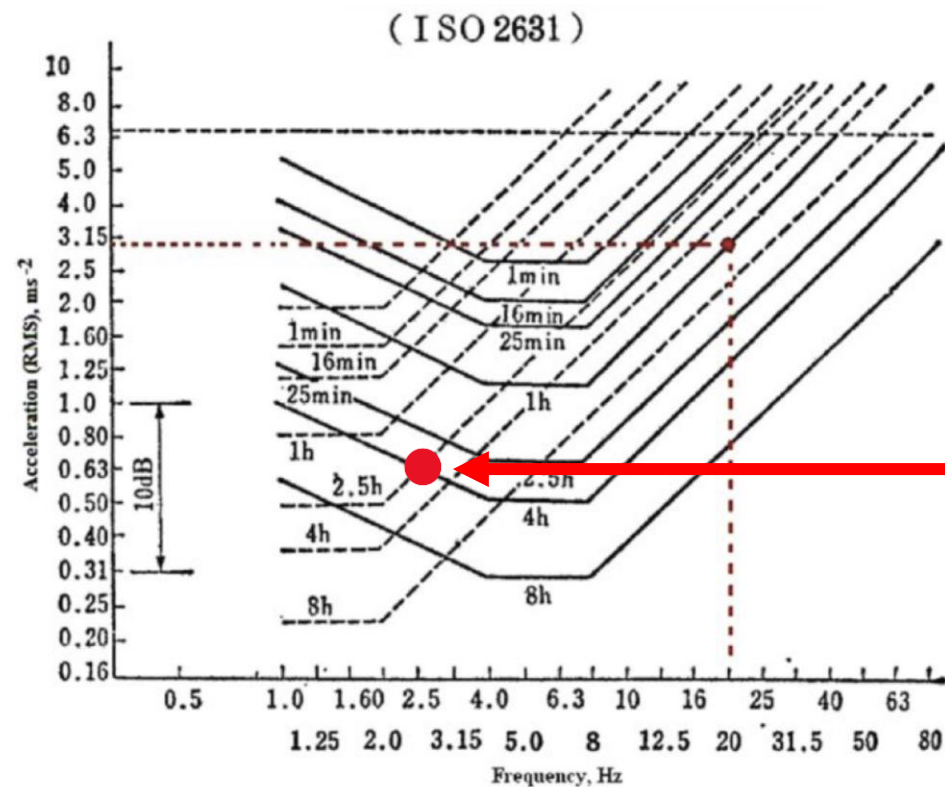
Temporal factor :

$$\theta_s(t) = \frac{\omega_s \sin \omega t - \omega \sin \omega_s t}{\omega_s (\omega_s^2 - \omega^2)}$$

Comfort level

RMS value : $\sqrt{f^2} = \frac{A}{\sqrt{2}} = 0,6560\text{m/s}^2$

ISO2631 :



Damping ratios

From $\mathbf{x}_{(r)}^T \mathbf{C} \mathbf{x}_{(r)}$, with $\mathbf{C} = \alpha \mathbf{K} + \beta \mathbf{M}$, the damping ratio expression is obtained :

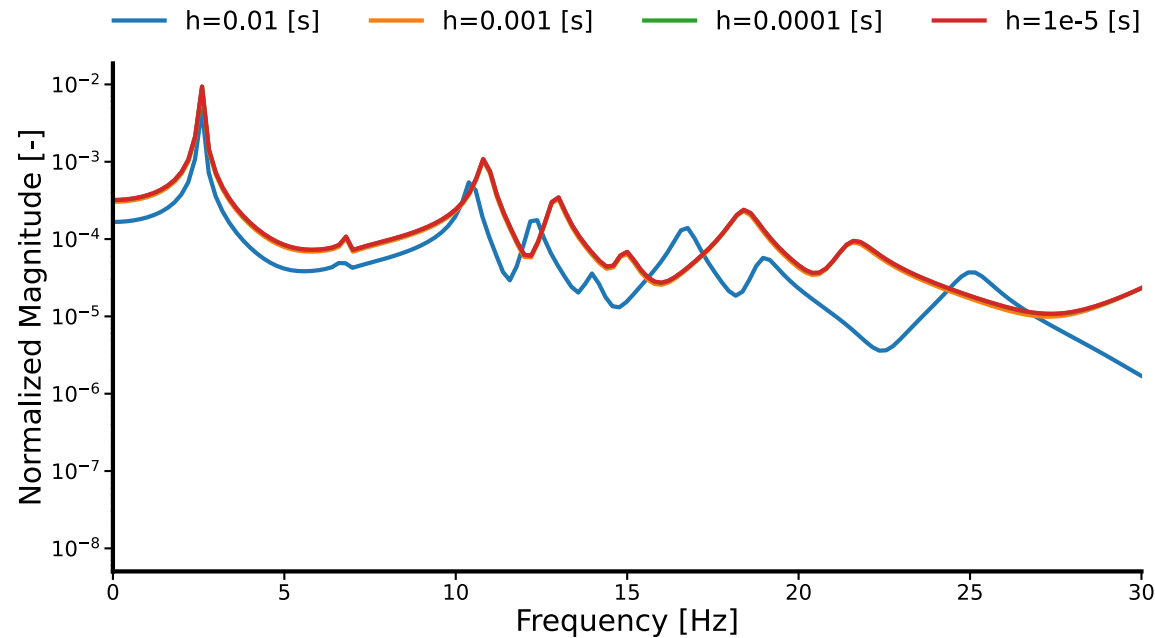
$$\varepsilon_{(r)} = \frac{1}{2} \left(\alpha \omega_{0r} + \frac{\beta}{\omega_{0r}} \right)$$

For the first six modes :

Mode [-]	1	2	3	4	5	6
Damping ratio [-]	0,010	0,010	0,013	0,015	0,016	0,017

Newmark algorithm : Integration parameters

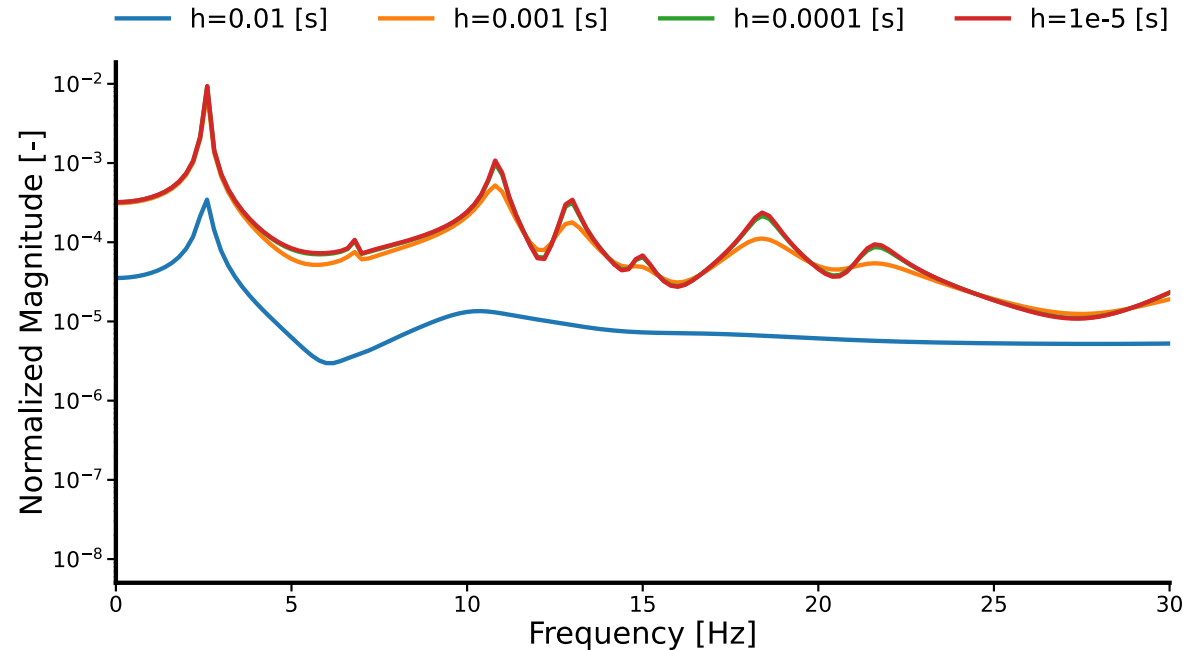
$$\underline{\gamma = 0,5, \beta = 0,25}$$



$$\frac{\Delta T}{T} = \frac{1}{2} \left(\beta + \frac{11}{48} + \frac{\gamma}{4} (\gamma - 3) \right) \omega^2 h^2 + O(\omega^4 h^4)$$

Newmark algorithm : Integration parameters

$$\gamma = 0,91, \beta = 0,5$$



$$\bar{\varepsilon} = \frac{1}{2} \left(\gamma - \frac{1}{2} \right) \omega h - \frac{1}{2} \left(\beta + \frac{\gamma}{2} - \frac{1}{4} \right) \left(\gamma - \frac{1}{2} \right) \omega^3 h^3 + O(\omega^5 h^5)$$

Reduction methods

Reduction matrix **R** given by

$$\mathbf{x} = \mathbf{R}\mathbf{x}_R$$

Guyan-Irons (static condensation):

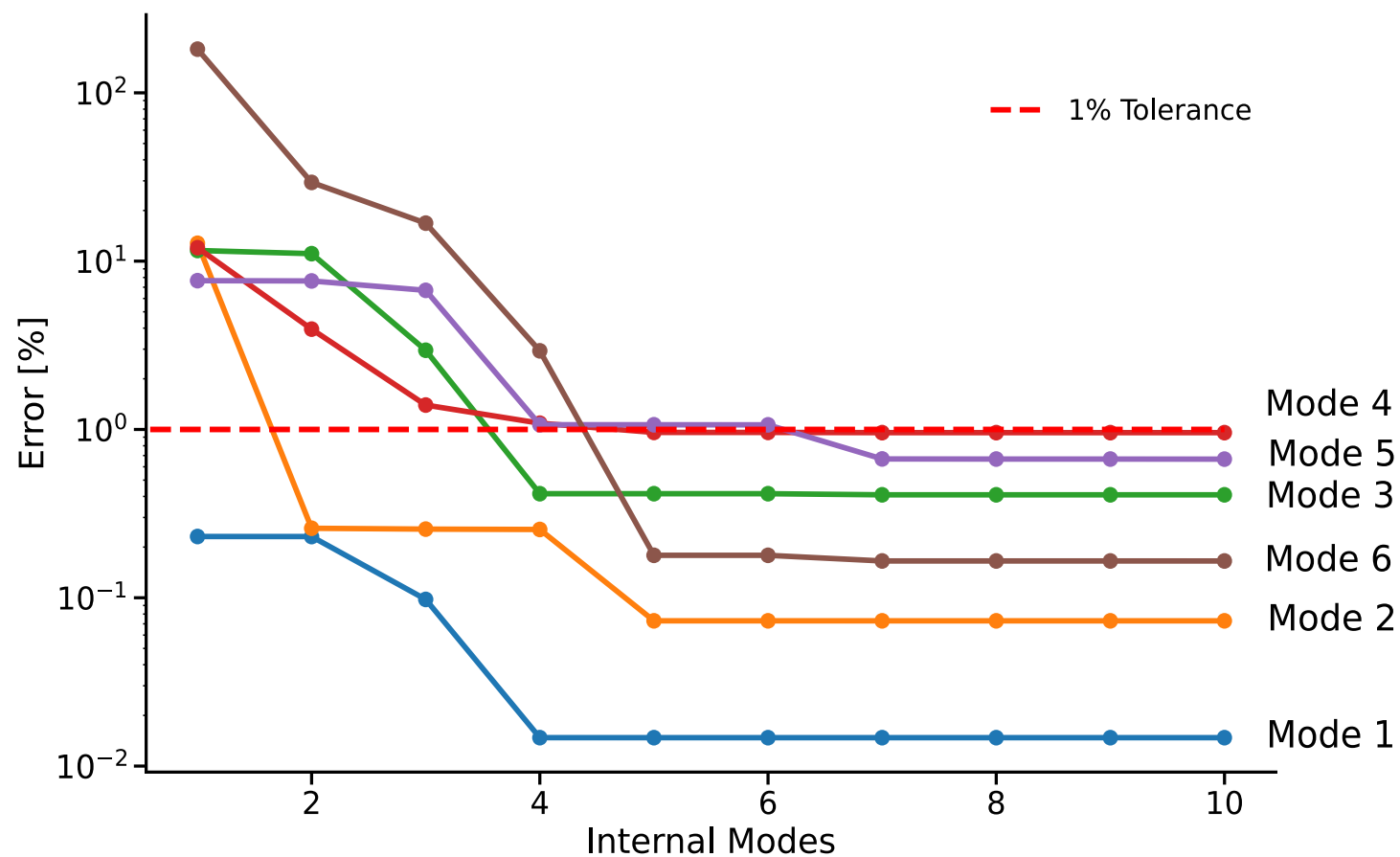
$$\mathbf{R} = \begin{pmatrix} \mathbf{I} \\ -\mathbf{K}_{CC}^{-1}\mathbf{K}_{CR} \end{pmatrix}$$

Craig-Bampton :

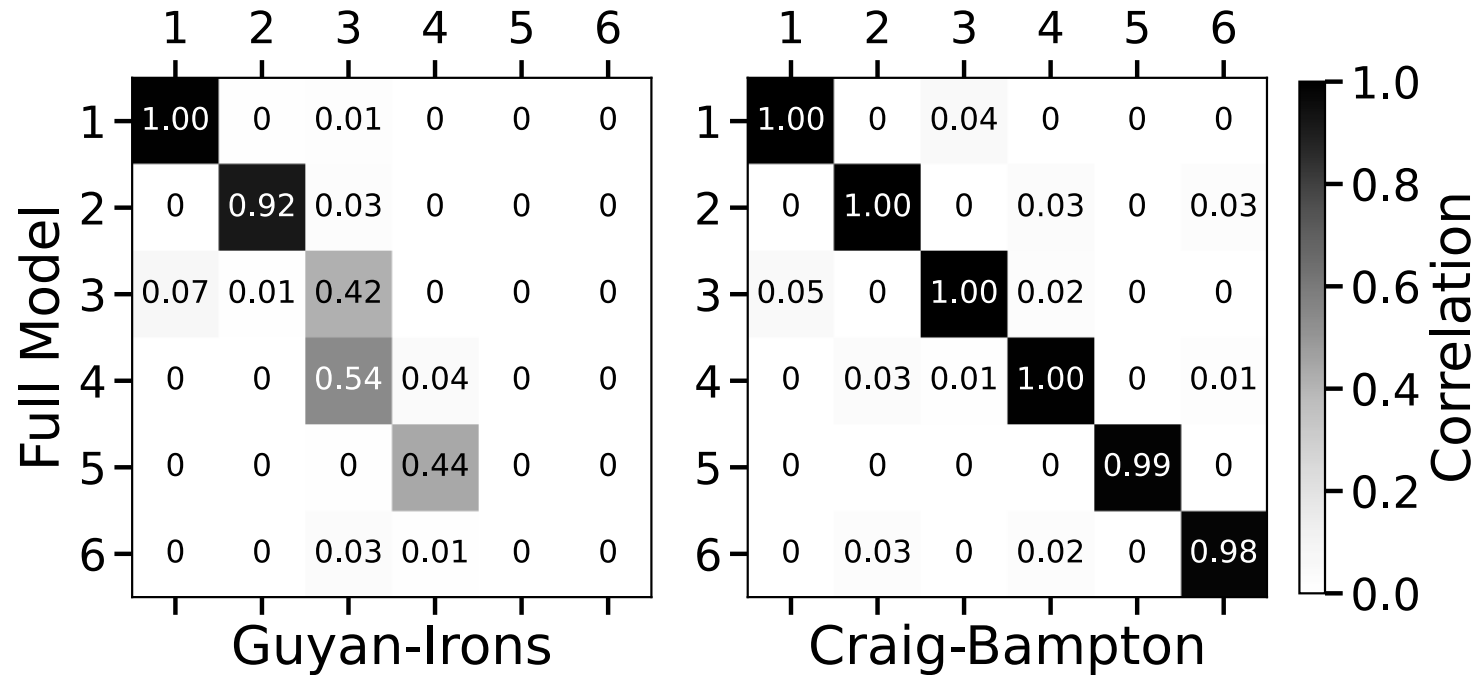
$$\mathbf{R} = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{K}_{II}^{-1}\mathbf{K}_{IB} & \Phi_n \end{pmatrix}$$

 Adjustable accuracy

Craig-Bampton accuracy

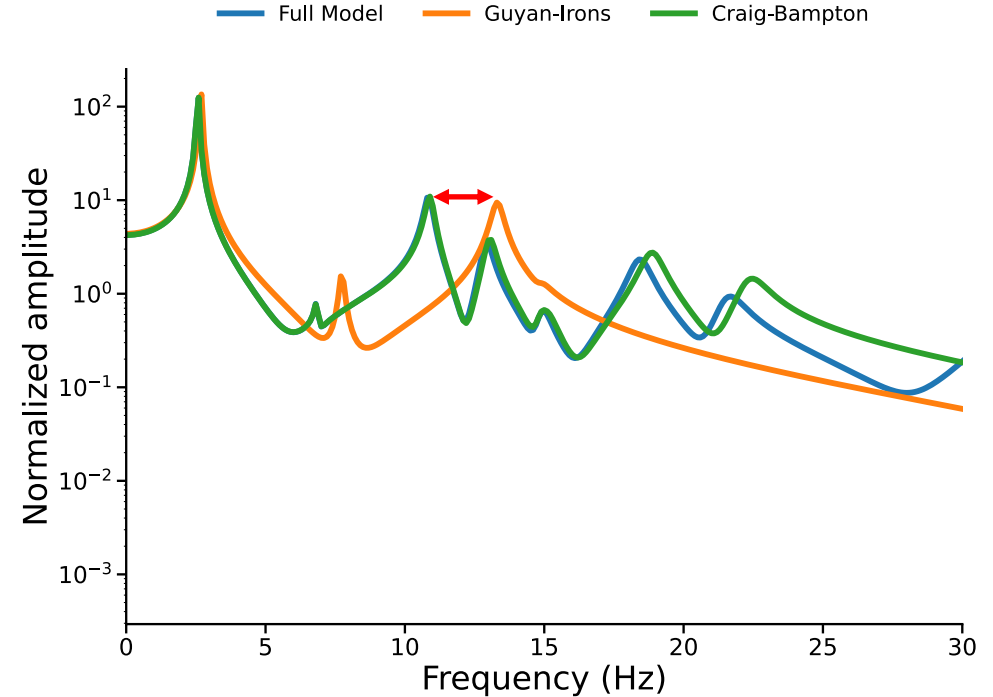
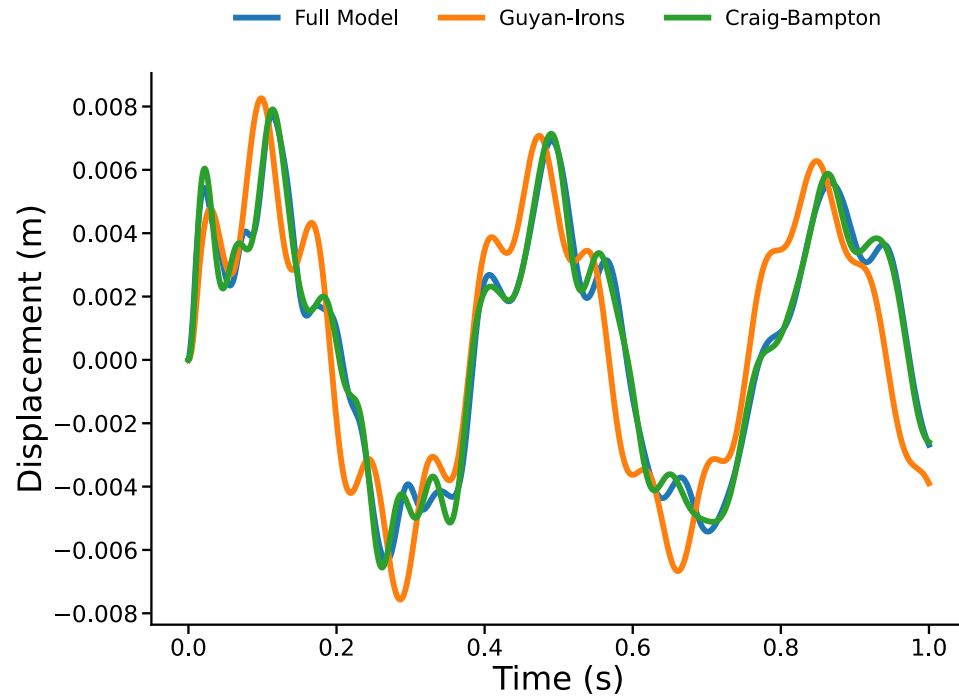


Modal Analysis Comparison

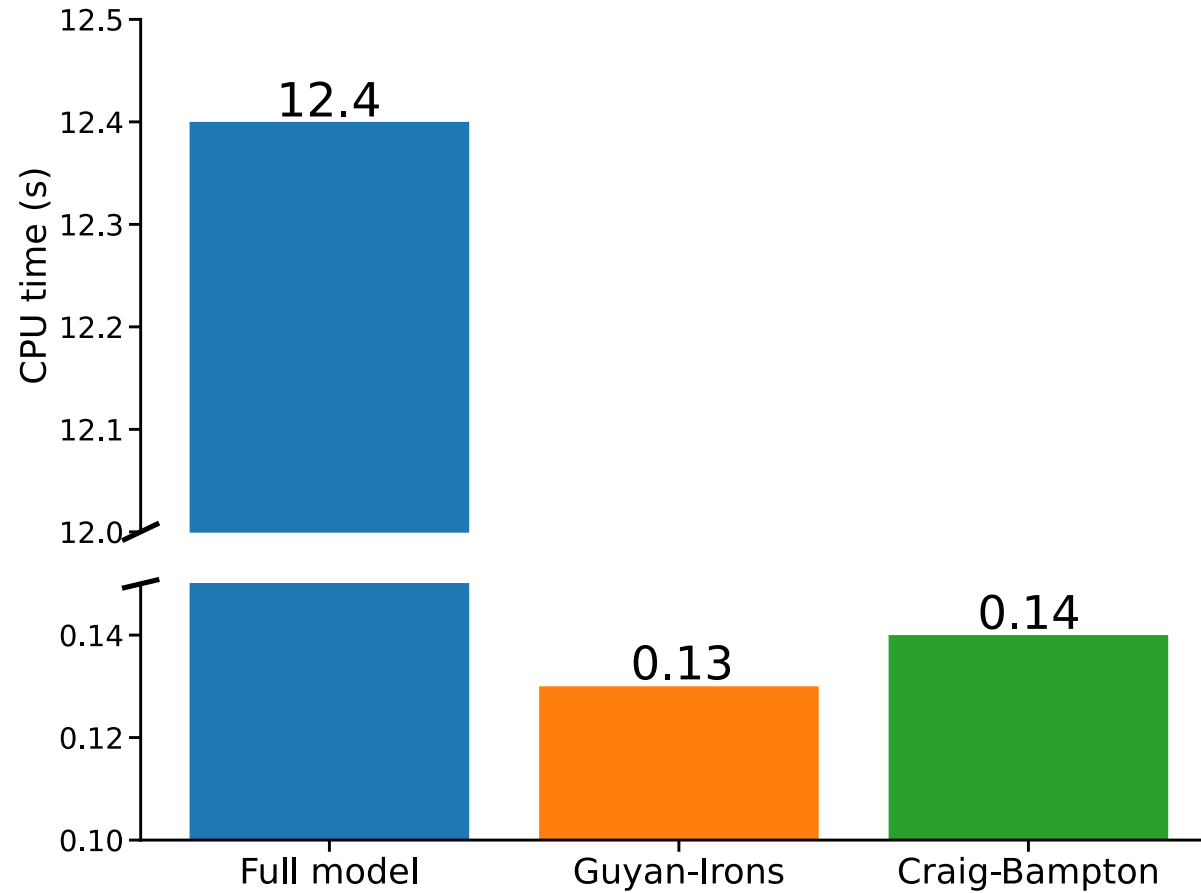


Relative error (%)	Mode 1	Mode 2	Mode 3	Mode 4	Mode 5	Mode 6
C-B (7 modes)	0.015	0.073	0.409	0.957	0.668	0.165
G-I	3.79	13.05	22.69	15.22	202.29	427.48

Transient Response Comparison



Computational Cost comparison



Conclusion

- **Modeling** : Convergence in 5 elements/beam.
Python stiffer than NX (Euler-Bernoulli).
- **Approximation methods** : MA method has better accuracy with a small number of modes. Performs best for smaller cost.
- **Newmark algorithm** : average constant acceleration scheme performs best.
Smaller time step = better the accuracy but trade-off with computation time.
- **Reduction methods** : C-B superior in accuracy and “adjustable”, small additional cost.